Final1sub.1

2026-10-23

R Markdown

```
#1- loading data, look at data set up way to clean
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
covar <- read_csv("GSE85307_covar-1.csv")
```

```
## Rows: 157 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (10): subjectID, geoID, mat_age_yrs, preg_preeclampsia, delivery_gestAge...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
expr  <- read_csv("GSE85307_geneExpression.csv")
```

```
## New names:
## Rows: 500 Columns: 158
## -- Column specification
## -------------------------------------------------------- Delimiter: "," dbl
## (158): ...1, GSM2264363, GSM2264379, GSM2264394, GSM2264418, GSM2264449,...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```
gene <- read_csv("GSE85307_geneNames.csv")
```

```
## Rows: 750 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): GENE
## dbl (1): PROBEID
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#view(covar)
#view(expr)
#view(gene)


##first col of expr has empty name, covar: description with data, remove those
##and better label

##expr first col rename to ProbID
expr <- expr %>% rename(ProbID = 1)

##Cleaning covar so that there is no description, col have right value type,
clean_covar <- covar %>%
  mutate(across(-c(subjectID, geoID), ~ str_remove(.x, "^.*?:\\s*"))) %>%
  mutate(
    mat_age_yrs = as.numeric(mat_age_yrs),
    delivery_gestAge_wks = as.numeric(delivery_gestAge_wks),
    sampleCollection_gestAge_wks = as.numeric(sampleCollection_gestAge_wks),
    mat_BMI = as.numeric(mat_BMI),
    mat_vitD = as.numeric(mat_vitD),
    preg_preeclampsia = factor(preg_preeclampsia, levels = c("Normal",
                                                             "Preeclampsia")),
    mat_asthma = factor(mat_asthma, levels = c("0", "1"),labels = c("Healthy",
                          "Asthma")), batch_Microarray = factor(batch_Microarray)
  ) %>%

## Gene:TMEM176B
  #Continuous covariate: VitD
  #Categorical 1: age_group
  #Categorical 2: preeclampsia
  # Derive a categorical variable from a continuous one (Age-group/maternal age)
  mutate(age_group = factor(
  ifelse(mat_age_yrs < 25, "<25",
         ifelse(mat_age_yrs < 30, "25-29", "30+")),
  levels = c("<25", "25-29", "30+")))

#view(clean_covar)
#connect probID to gene so that I can now specify my gene of interest

## expression file has probe ID but no gene names, gene file pairs probe ID
##with gene name. Use left join to combine these two tables. The column names
##are different so have to tell r they are actually the same info. To make
##easier to read use relocate to move the new gene column next to probe id.
##Check to make sure every probe found a match
```

```r
expr_named <- expr %>%
  left_join(gene, by =c("ProbID" = "PROBEID")) %>%
  relocate(GENE, .after = ProbID)
sum(is.na(expr_named$GENE))
```

```
## [1] 0
```

```r
#assigning my gene of interest
my_gene <- "TMEM176B"


##I have my gene now I need to figure out where it is in the table. reshape
##table into two columns that I can merge on my covariates. Dont need the
##probID or gene columns. I also want my gene names to be rows so I have to
##pivot_longer.

gene_long <- expr_named %>%
  filter(GENE == my_gene) %>%
  select(-ProbID, -GENE) %>%
  pivot_longer(everything(), names_to = "geoID", values_to = "expression")

#save pivot table to dat
dat <- gene_long %>%
  inner_join(clean_covar, by = "geoID")

## Checking data to make sure nothing was lost and everything matches correctly.

## Count the rows in merged data, does sample Id in covariate file appear as a
## col name in expression file, does every sample col in expression file appear
## in covariate file (except for first column), ask if expression values are
## missing, and count how many samples fall into each combination of
## preeclampsia status and age group

nrow(dat)
```

```
## [1] 157
```

```r
all(clean_covar$geoID %in% names(expr))
```

```
## [1] TRUE
```

```r
all(names(expr)[-1] %in% clean_covar$geoID)
```

```
## [1] TRUE
```

```r
anyNA(dat$expression)
```

```
## [1] FALSE
```

```r
table(dat$preg_preeclampsia, dat$age_group)
```

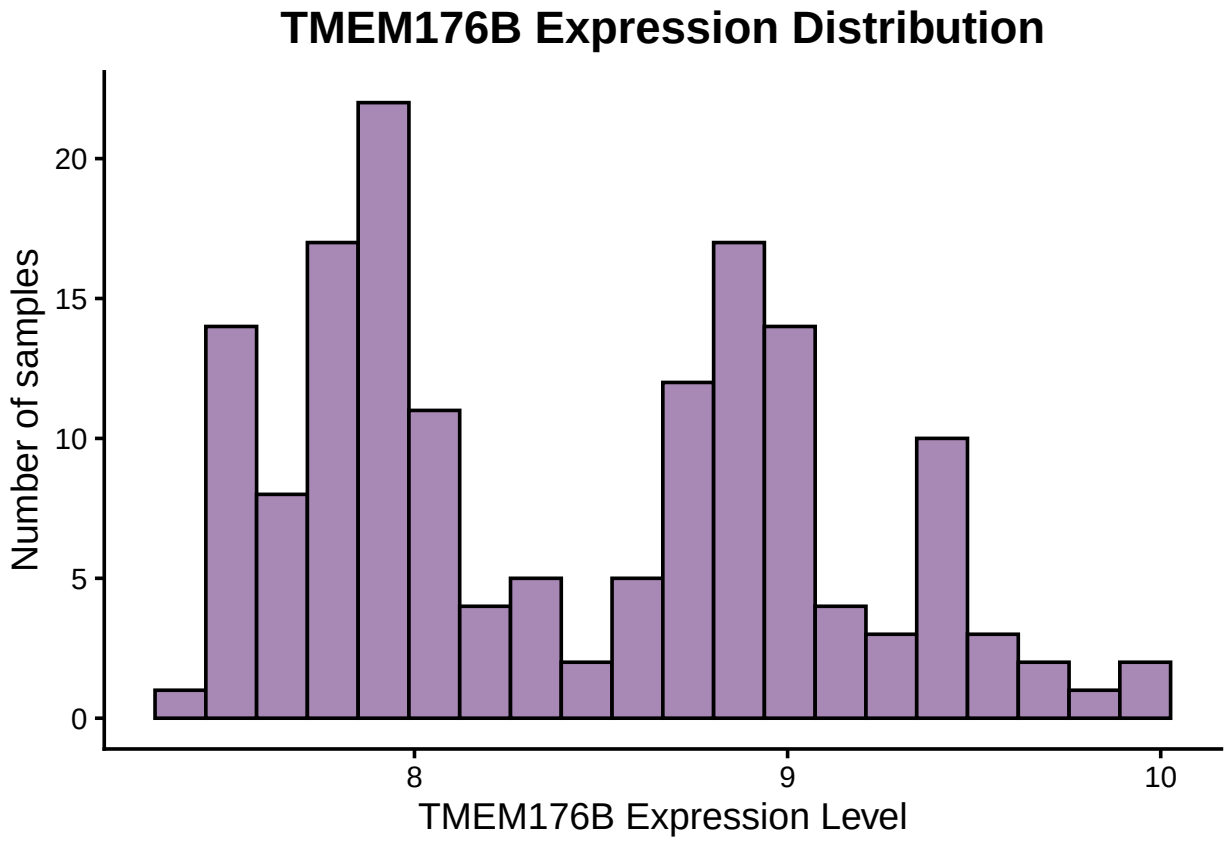
```
##
##                <25 25-29 30+
##   Normal        46    34  30
##   Preeclampsia  22    15  10
```

```r
#Setting theme and colors
theme_1 <- theme_classic(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
        legend.position = "right")

pe_colors <- c("Normal" = "#A888B5", "Preeclampsia" = "#EC5353")


#Histogram of gene expression
p_hist <- ggplot(dat, aes(x = expression)) +
  geom_histogram(bins = 20, fill = "#A888B5", color = "black") +
  labs(title = paste(my_gene, "Expression Distribution"),
       x = paste(my_gene, "Expression Level"),
       y = "Number of samples") +
  theme_1

p_hist
```

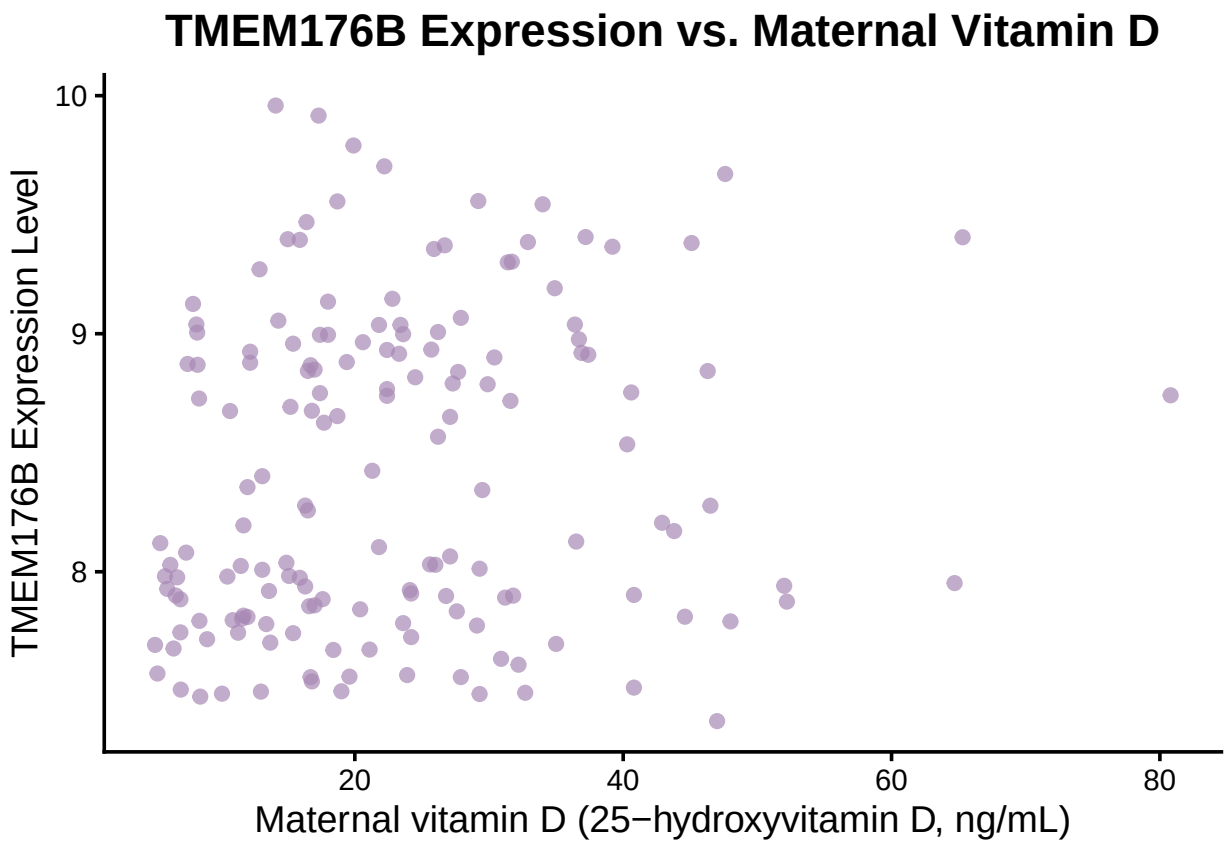


```r
#saving plot as pdf for presentations
ggsave("sub1_plots_hist1.pdf")
```

```
## Saving 6.5 x 4.5 in image
```

```r
# scatter plot vitD vs expression
p_scatter <- ggplot(dat, aes(x = mat_vitD, y = expression)) +
  geom_point(alpha = 0.7, size = 2, color = "#A888B5") +
  labs(title = paste(my_gene, "Expression vs. Maternal Vitamin D"),
       x = "Maternal vitamin D (25-hydroxyvitamin D, ng/mL)",
       y = paste(my_gene, "Expression Level")) +
  theme_1

p_scatter
```



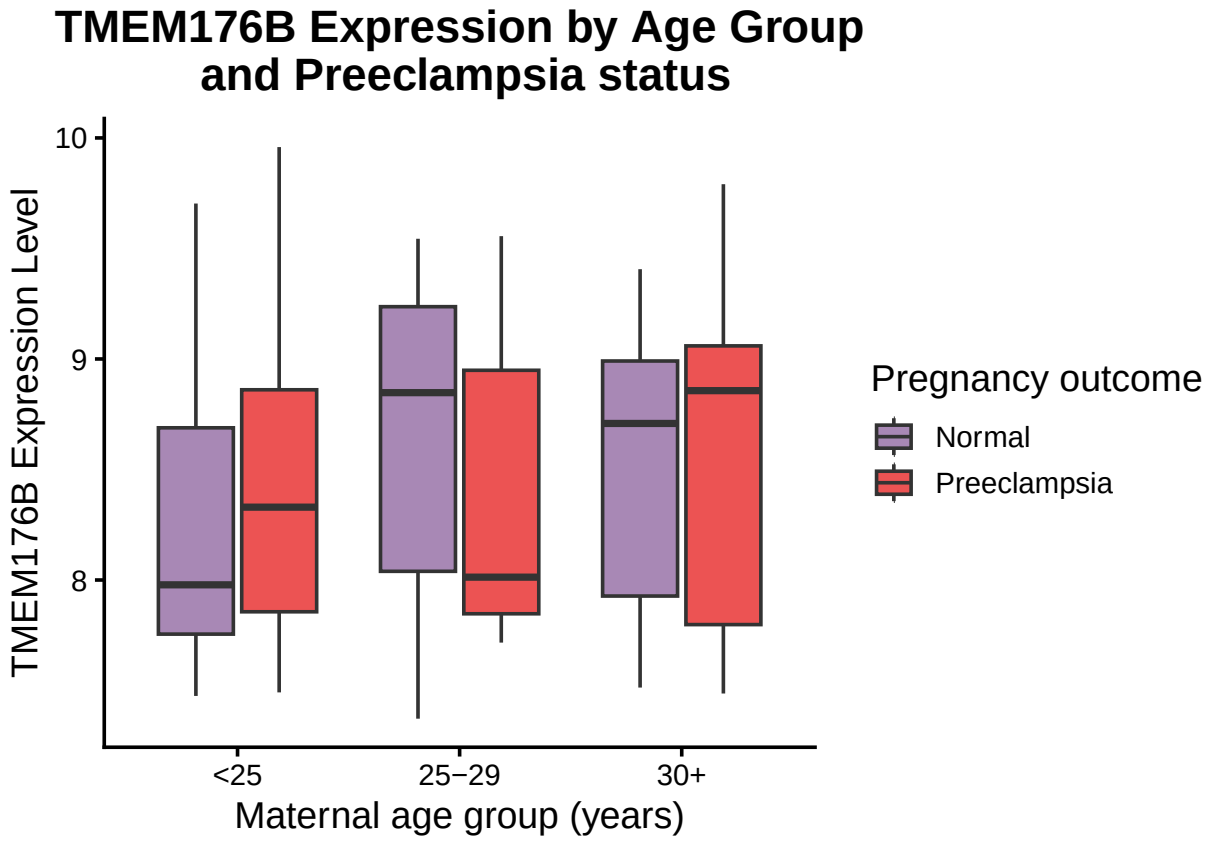

```r
#saving plot as pdf for presentations
ggsave("sub1_plots_scatter_1.pdf")
```

```
## Saving 6.5 x 4.5 in image
```

```r
# Boxplot preeclampsia vs age group
p_box <- ggplot(dat, aes(x = age_group, y = expression,
                         fill = preg_preeclampsia)) +
  geom_boxplot() +
```

```
  scale_fill_manual(values = pe_colors, name = "Pregnancy outcome") +
  labs(title = paste(my_gene,
                     "Expression by Age Group\n and Preeclampsia status"),
       x = "Maternal age group (years)",
       y = paste(my_gene, "Expression Level")) +
  theme_1

p_box
```




```
#saving plot as pdf for presentations
ggsave("sub1_plots.pdf")
```

```
## Saving 6.5 x 4.5 in image
```

```
# used material from other classes (Data wrangling and Biostat) and AI for debugging
```

```
#Start of submission2

#need somewhere to store my genes
my_genes <- c("TMEM176B", "UTS2", "CLIC2")

#need a table with one row per sample and one column per gene.
genes_wide <- expr_named %>%
  filter(GENE %in% my_genes)%>%
```

```r
  select(-ProbID)%>%
  pivot_longer(-GENE, names_to = "geoID", values_to = "expression") %>%
  pivot_wider(names_from = GENE, values_from = expression)

#each gene name becomes its own column,filled with the genes expression values.(merging genes_wide with

dat_all <- genes_wide %>%
  inner_join(clean_covar, by = "geoID")

#check to make sure it includes genes (it is)
##nrow(dat_all)
##names(dat_all)

#make theme for plots based on prof.comment in submission 1 (added more x/y ticks in histogram and scat
  legible_theme <- theme_minimal(base_size = 14) +
    theme(
      plot.title   = element_text(size = 16, face = "bold"),
      axis.title   = element_text(size = 14),
      axis.text    = element_text(size = 12),
      legend.title = element_text(size = 13),
      legend.text  = element_text(size = 12),
    )



#making function:
# wants function to make histogram, scatter plot and boxplot for each gene. for each gene in list, copy


make_gene_plots <- function(df, genes) {


  for (g in genes) {

    # copy this gene's values into a column called "expression"
    df$expression <- df[[g]]

    # Histogram of expression
    print(ggplot(df, aes(x = expression)) +
      geom_histogram(bins = 20, fill = "#A888B5", color = "white") +
      labs(title = paste(g, "Expression Distribution"),
           x = paste(g, "Expression Level"),
           y = "Number of samples")+
        scale_x_continuous(n.breaks = 10) +
        scale_y_continuous(n.breaks = 10) +
        legible_theme)

    # Scatterplot: vitamin D vs expression
    print(ggplot(df, aes(x = mat_vitD, y = expression)) +
      geom_point(alpha = 0.7, size = 2, color = "#A888B5") +
      labs(title = str_wrap(paste(g, "Expression vs. Maternal VitaminD"), width = 40),
           x = "Maternal vitamin D (25-hydroxyvitamin D, ng/mL)",
           y = paste(g, "Expression Level"))+
```

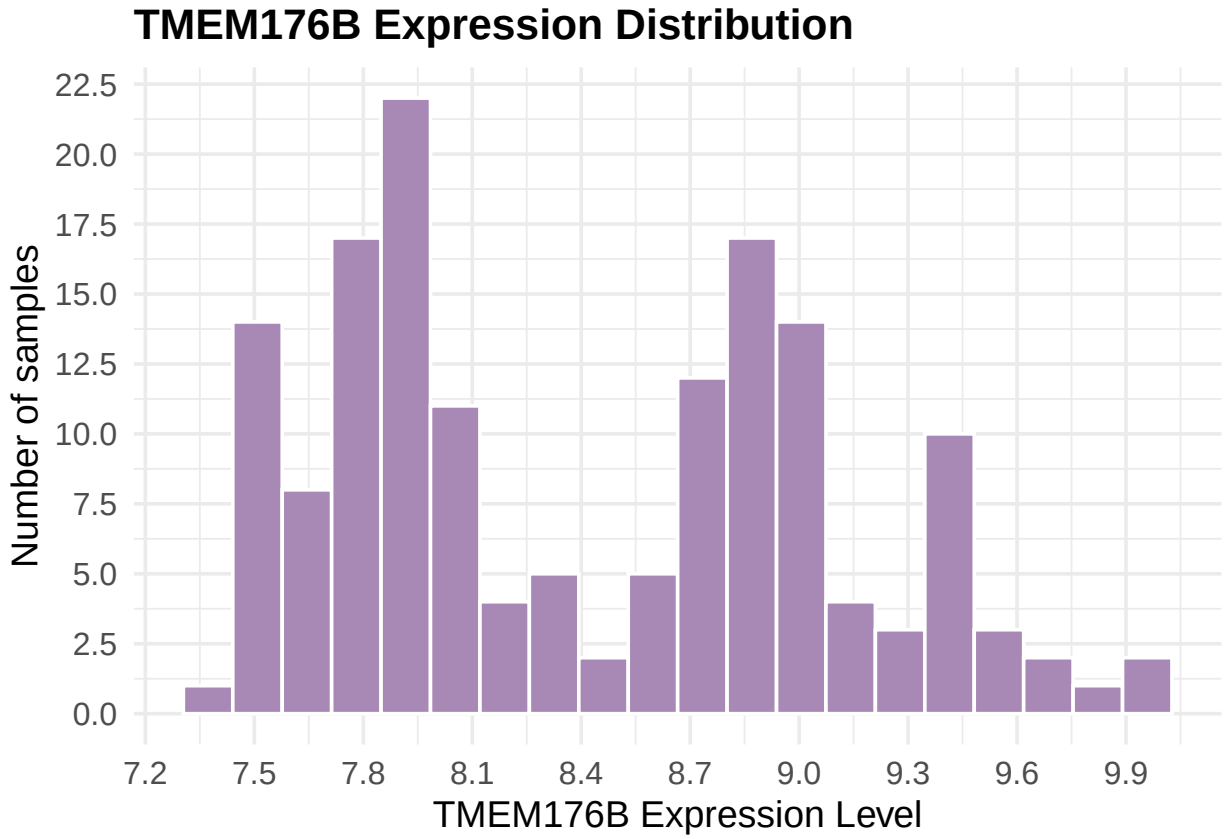
```
        scale_x_continuous(n.breaks = 10) +
        scale_y_continuous(n.breaks = 10) +
        legible_theme)

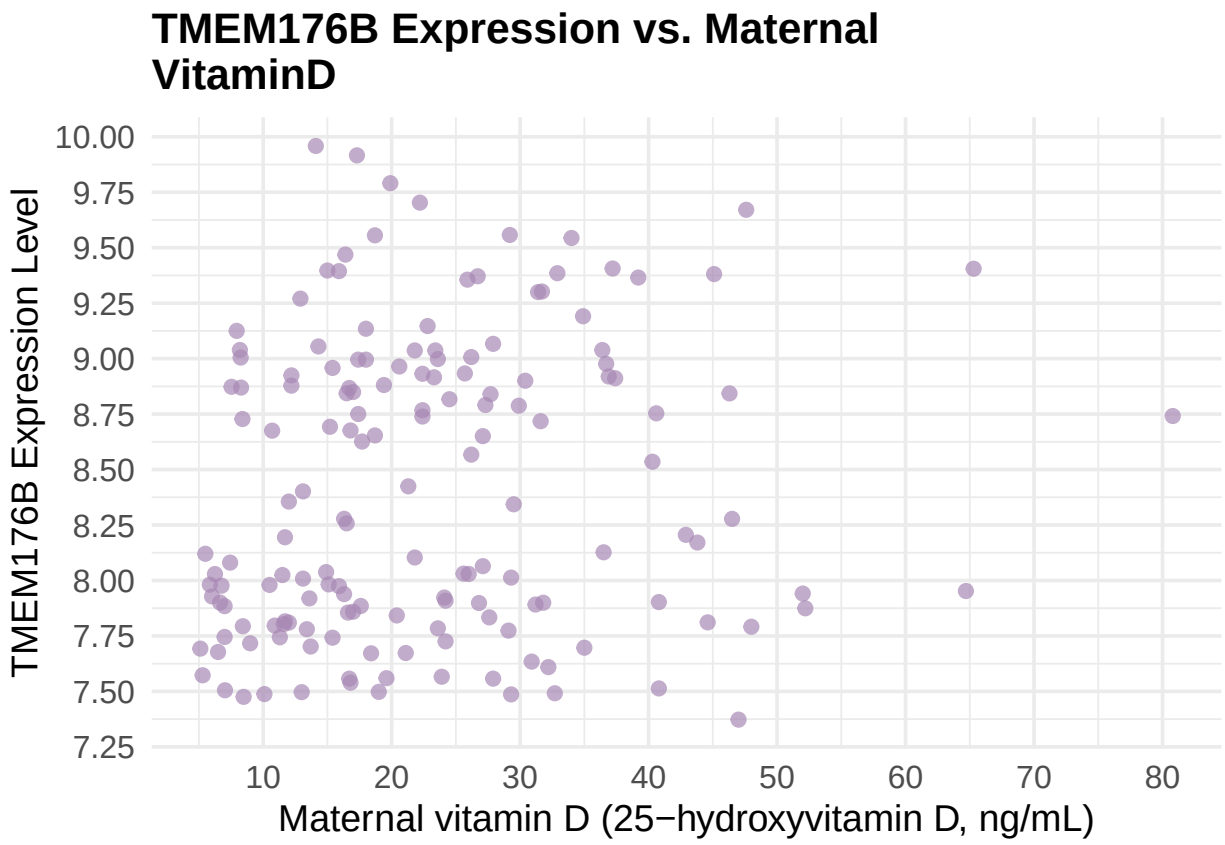
    # Boxplot: age group and preeclampsia status
    print(ggplot(df, aes(x = age_group, y = expression, fill = preg_preeclampsia)) +
      geom_boxplot() +
      scale_fill_manual(values = c("#A888B5", "#EC5353"), name = "Pregnancy outcome") +
      labs(title = str_wrap(paste(g, "Expression by Age Group and Preeclampsia Status"),
                            width = 40),
           x = "Maternal age group (years)",
           y = paste(g, "Expression Level"))+
        # tried to do geom_jitter but it didn't look right, so i had to look up an alternative
       geom_point(position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.8),
             size = 1.5, alpha = 0.7, show.legend = FALSE)+
        legible_theme)

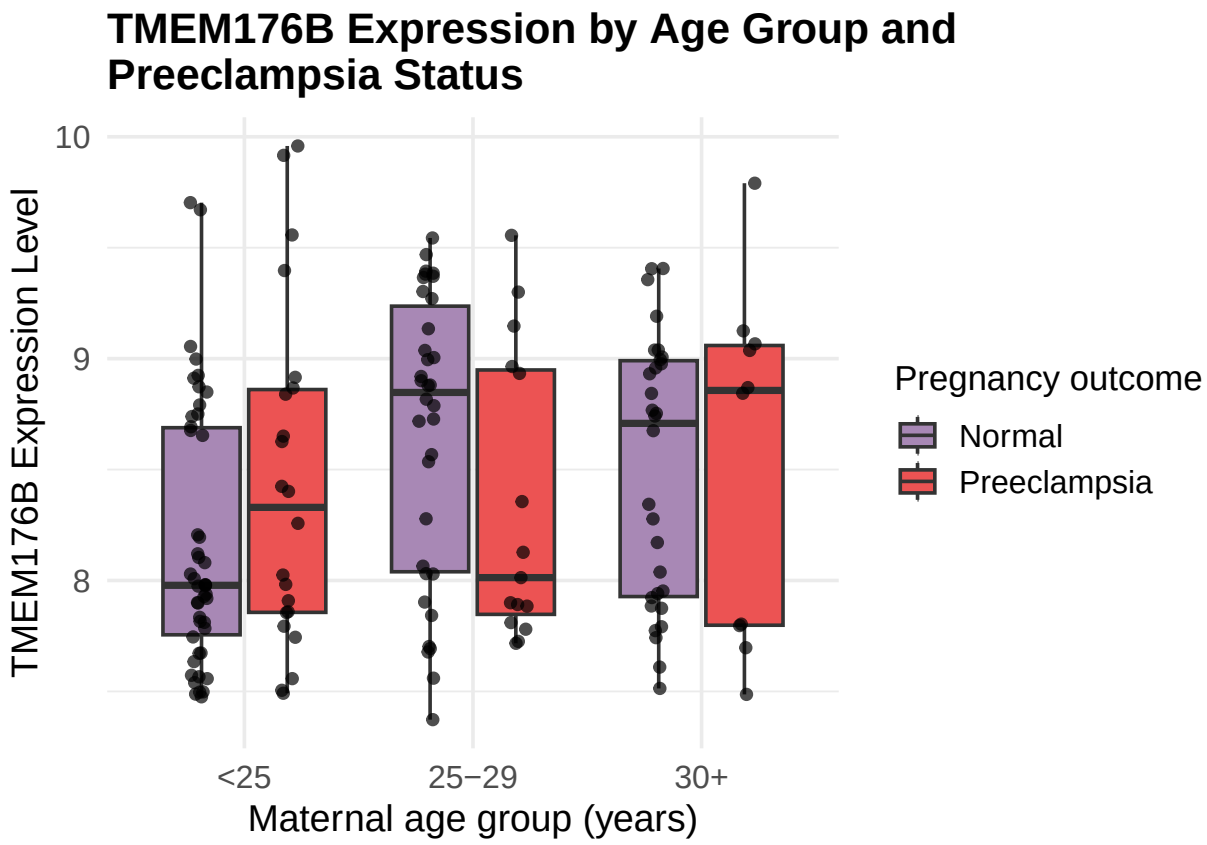
  }

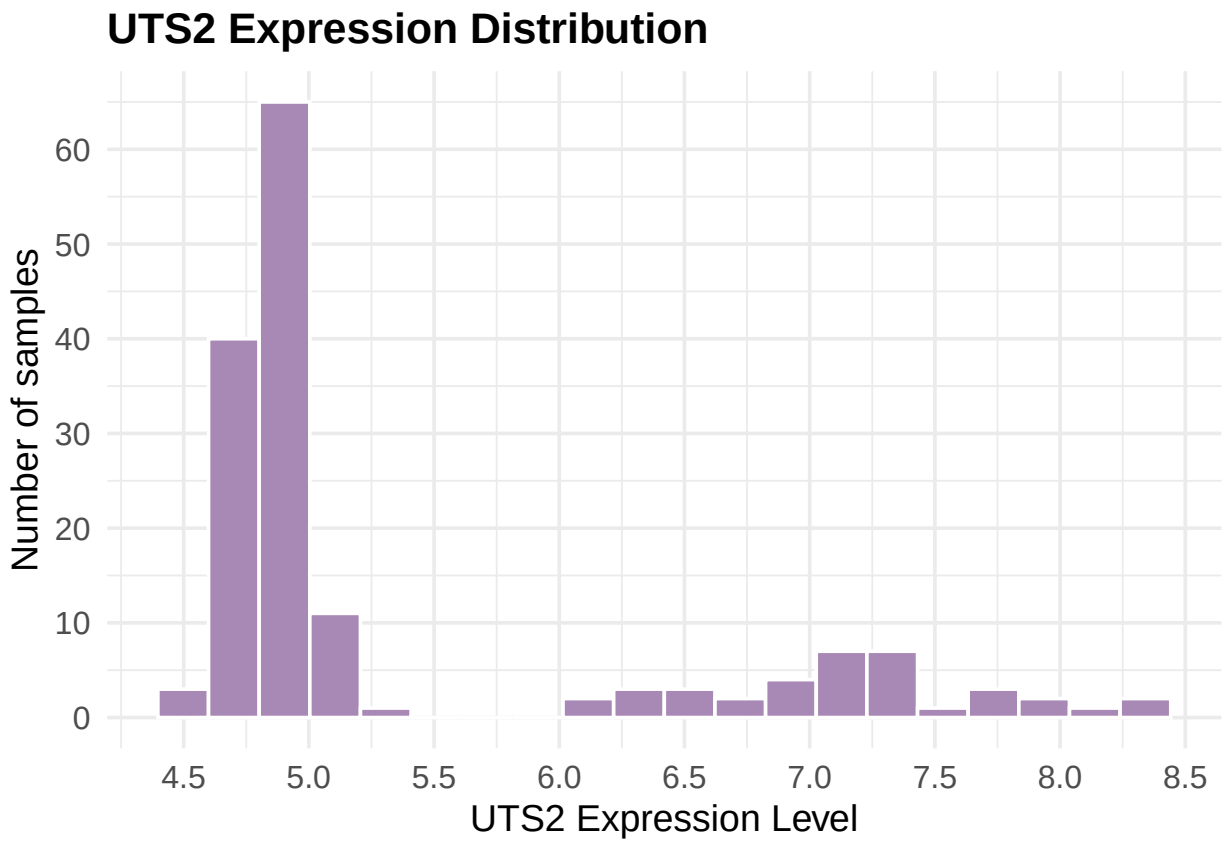
}

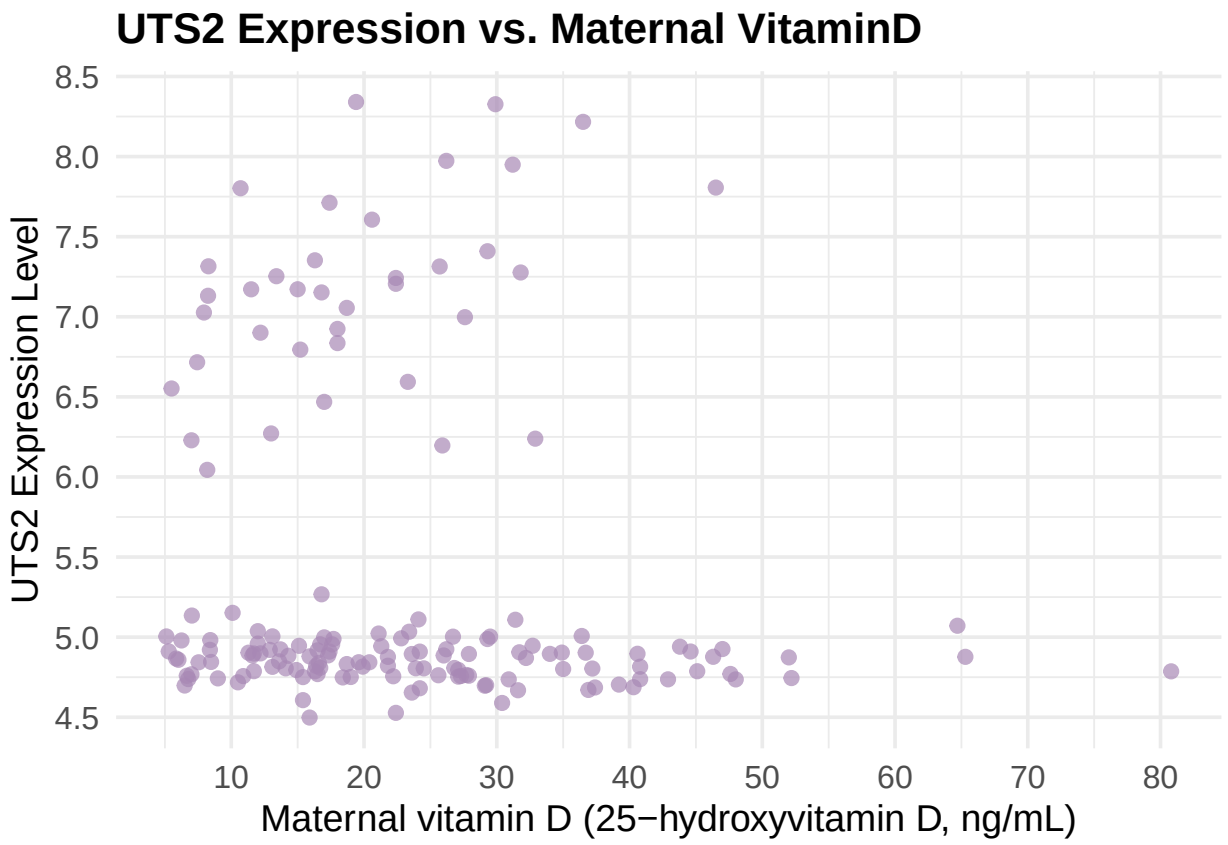
# call it
make_gene_plots(dat_all, my_genes)
```

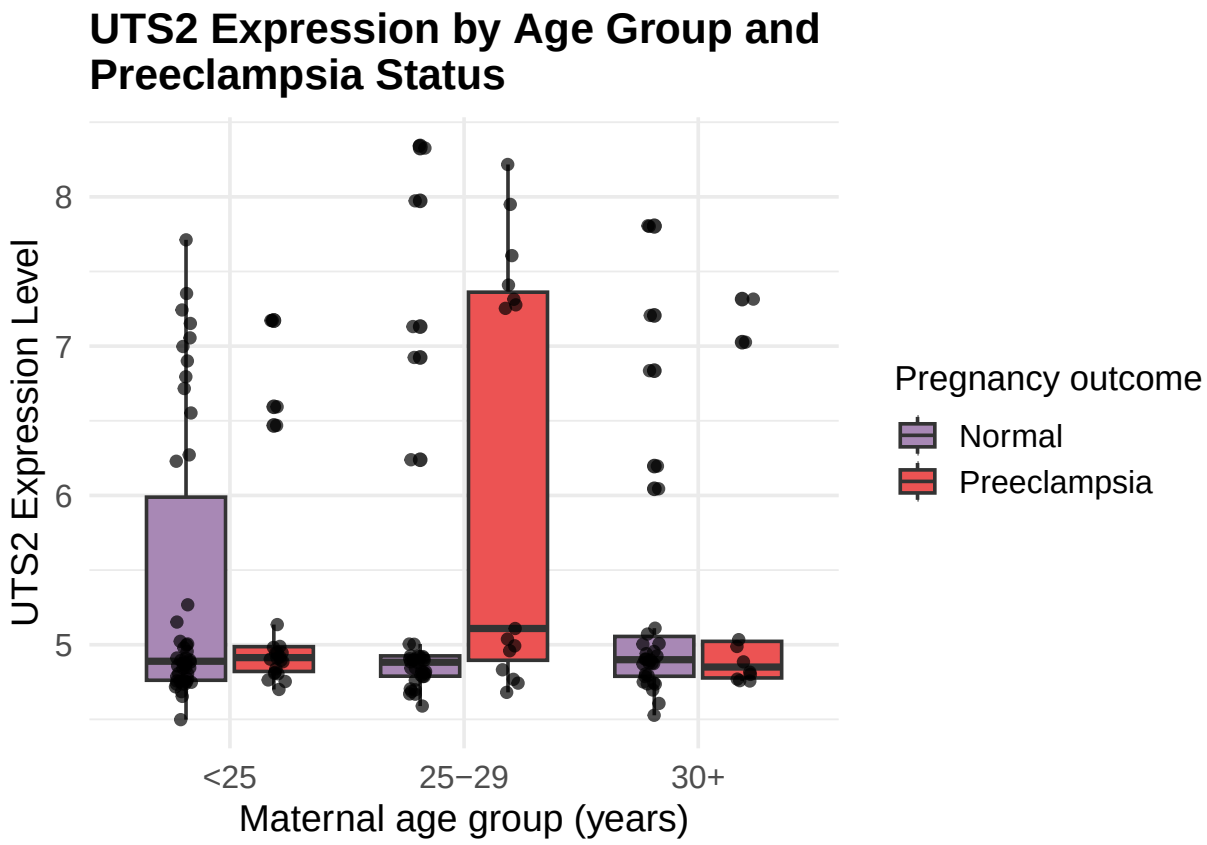


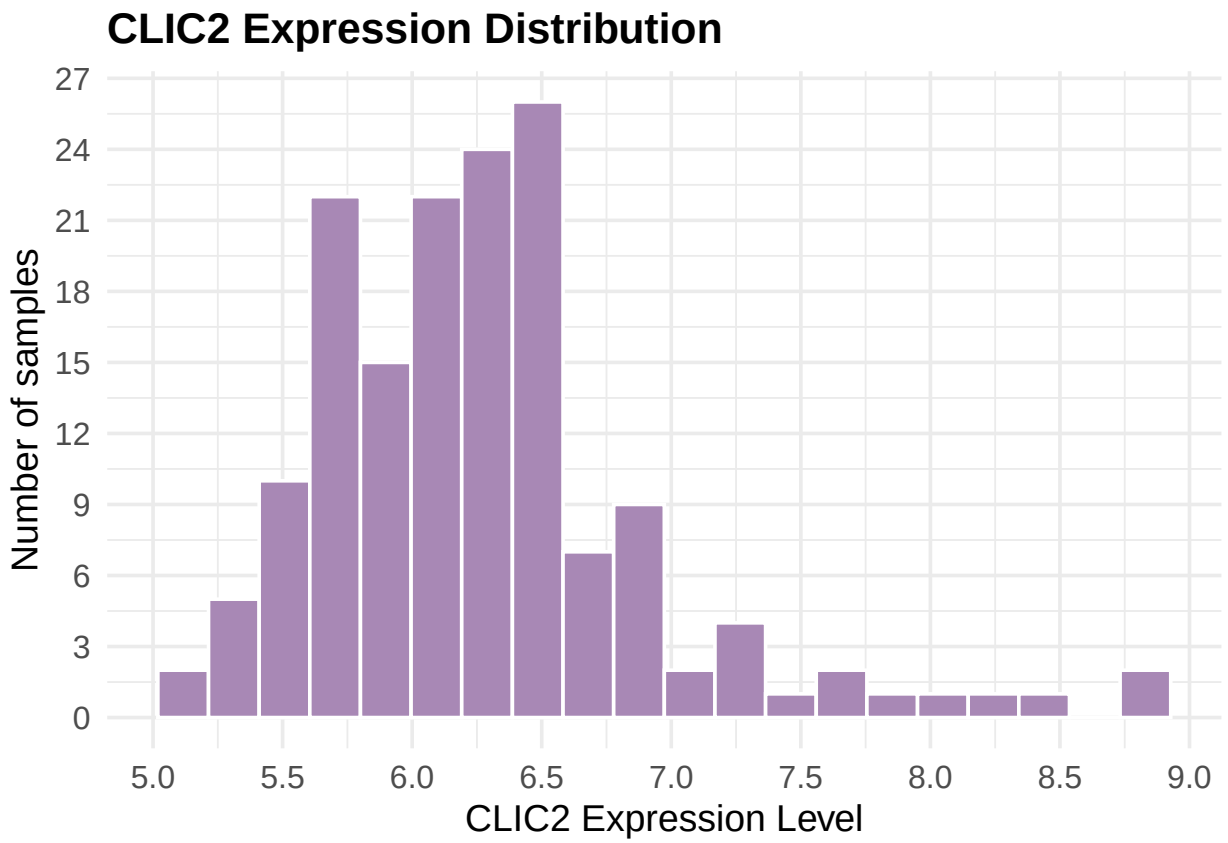

TMEM176B Expression vs. Maternal VitaminD
TMEM176B Expression Level
10.00
9.75
9.50
9.25
9.00
8.75
8.50
8.25
8.00
7.75
7.50
7.25
10
20
30
40
50
60
70
80
Maternal vitamin D (25−hydroxyvitamin D, ng/mL)

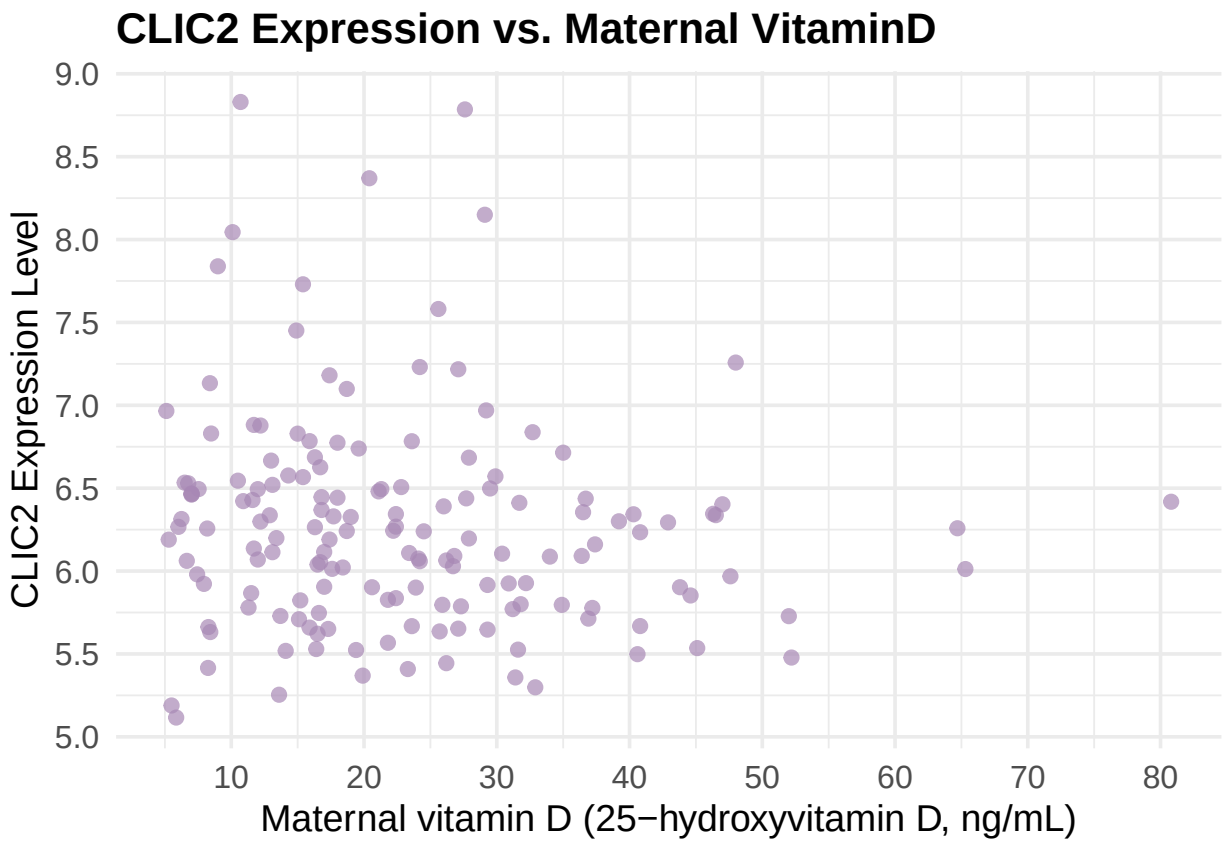

TMEM176B Expression by Age Group and Preeclampsia Status
TMEM176B Expression Level
10
9
8
<25
25−29
30+
Maternal age group (years)
Pregnancy outcome
Normal
Preeclampsia

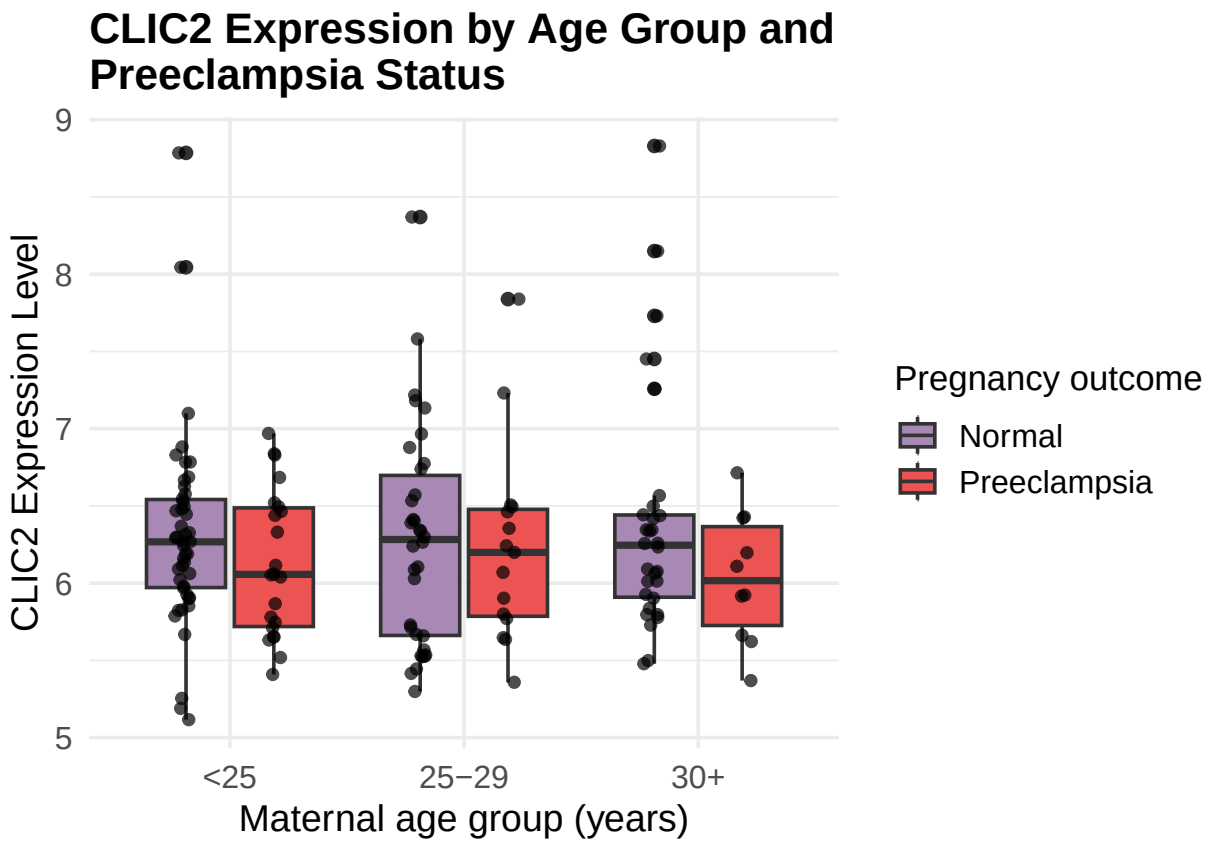

UTS2 Expression Distribution
Number of samples
0
10
20
30
40
50
60
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
8.5
UTS2 Expression Level


UTS2 Expression vs. Maternal VitaminD
UTS2 Expression Level
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
4.5
10
20
30
40
50
60
70
80
Maternal vitamin D (25−hydroxyvitamin D, ng/mL)


UTS2 Expression by Age Group and Preeclampsia Status
UTS2 Expression Level
8
7
6
5
<25
25−29
30+
Maternal age group (years)
Pregnancy outcome
Normal
Preeclampsia


CLIC2 Expression Distribution
Number of samples
0
3
6
9
12
15
18
21
24
27
5.0
5.5
6.0
6.5
7.0
7.5
8.0
8.5
9.0
CLIC2 Expression Level


CLIC2 Expression vs. Maternal VitaminD
CLIC2 Expression Level
9.0
8.5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
10
20
30
40
50
60
70
80
Maternal vitamin D (25−hydroxyvitamin D, ng/mL)




```
#used AI and google to look specific things up that I forgot or to debugg.
```